

MARVIN JOSEPH B

APPLIED AI ENGINEER | LLM AGENTS | PRODUCTION AI

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PROFESSIONAL SUMMARY

Applied AI Engineer transitioning from 5+ years as a Senior Production SQL Server DBA, combining hands-on LLM and agentic application development with deep experience in mission-critical systems, incident response, reliability, and automation.

Hands-on experience building Python/FastAPI AI services, Pydantic-validated workflows, bounded LLM tool calling, evidence-grounded generation, AI evaluation, and human-in-the-loop controls. Combines deep production incident-response experience with Generative AI engineering, with emphasis on reliability, validation, least privilege, failure handling, and controlled execution.

TECHNICAL SKILLS

Generative AI & Agentic Systems:

Generative AI, LLM APIs, OpenAI Responses API, AI Agents, Agentic Workflows, Tool / Function Calling, Agent Orchestration, Structured Outputs, Evidence-Grounded Generation, RAG, Human-in-the-Loop (HITL), Prompt Engineering, Context Design, AI Evaluation

Backend & Software Engineering:

Python, FastAPI, Pydantic, REST APIs, PostgreSQL, SQL, Async Workflows, JSON, Schema Validation, Data Validation, Automated Testing, Git, GitHub, Pull Requests, Code Review

Frontend Engineering:

TypeScript, JavaScript, React, Node.js, Vite

DevOps & Production AI:

Docker, Docker Compose, Nginx, Linux, Ubuntu, GitHub Actions, CI/CD, Azure DevOps, AI Guardrails, Restricted Tool Access, Least-Privilege Design, Tool & Iteration Budgets, Human Approval Gates, Health Checks, Controlled Releases, Rollback

Database & Data Engineering:

SQL Server 2012–2022, T-SQL, PowerShell, Azure SQL, Always On Availability Groups, Query Store, Execution Plans, Dynamic Management Views, Wait Statistics, Backup & Recovery, SSIS, CDC, Pandas, ETL, P1/P2 Incident Response, Root-Cause Analysis, HA/DR

PROFESSIONAL EXPERIENCE

DIPLOMATIC SOLUTIONS CORPORATION — Frederick, MD

Senior Production SQL Server DBA | September 2022–Present

Internal Assignment: Applied AI Engineer — Incident Automation

- Built the SQL Server investigation path for an internal AI-assisted incident platform, translating production incident-response procedures into restricted Python diagnostics, FastAPI components, Pydantic evidence contracts, and bounded LLM tool-calling capabilities.
- Developed restricted diagnostic operations for SQL Server blocking, active requests, connection pressure, and database health, preventing the model from generating or executing arbitrary SQL.
- Designed Pydantic-validated evidence records with application-controlled identifiers and provenance, integrating them into the LLM workflow so generated reports could reference verified evidence instead of unsupported model claims.
- Integrated database diagnostics into a bounded AI agent workflow where the model selected approved tools while application code controlled execution, permissions, validation, usage limits, and failure handling.
- Created sanitized incident scenarios and AI evaluation tests covering tool selection, evidence completeness, citation validity, expected root-cause conclusions, invalid outputs, and failure conditions.
- Implemented safeguards including least-privileged access, strict tool schemas, tool and iteration limits, human approval gates, current-state revalidation, and post-action recovery verification.
- Developed and tested FastAPI investigation components through Git-based pull-request workflows with code review, CI validation, staging validation, controlled releases, and production troubleshooting.

Production Database Engineering

- Support 24x7 mission-critical SQL Server environments across on-premises and Azure infrastructure, including **50+ SQL Server instances, 200+ databases, and databases up to 4 TB.**
- Serve as a senior database escalation resource during P1/P2 incidents involving database failures, application connectivity, performance degradation, resource pressure, and service availability.
- Perform root-cause analysis for blocking chains, deadlocks, long-running queries, CPU and I/O pressure, TempDB contention, connection exhaustion, wait statistics, and execution-plan regressions.

- Administer Always On Availability Groups, backup and recovery, failover procedures, HA/DR configurations, database health monitoring, and production recovery operations.
 - Build PowerShell and T-SQL automation and support production changes through GitHub, Azure DevOps, CI/CD pipelines, and Octopus Deploy.
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AI ENGINEERING PROJECT

INCIDENT INVESTIGATION AGENT

Python | FastAPI | Pydantic | PostgreSQL | OpenAI Responses API | Docker | Nginx | GitHub Actions | React | TypeScript

Live Demo: marvinjb.dev/demo/incident-investigation

GitHub: github.com/marvinjbb

- Built and deployed a bounded AI agent that investigates controlled incidents, including blocked database queries, connection exhaustion, and failed application deployments, using restricted diagnostic tools.
 - Designed an application-controlled tool-calling loop where the LLM selects diagnostics while Python controls credentials, execution, permissions, tool access, iteration limits, evidence identity, and final report acceptance.
 - Implemented structured Pydantic-validated investigation reports with application-owned evidence IDs and deterministic citation validation so conclusions must reference evidence actually collected.
 - Separated investigation from remediation through deterministic policy checks, explicit human approval, proposal expiration, current-state revalidation, idempotent execution, and post-action recovery verification.
 - Added production controls, including session ownership, rate limiting, audit events, health checks, trusted-host policies, database migrations, structured logging, and restricted API access.
 - Containerized and deployed the FastAPI/PostgreSQL services on an Ubuntu VPS behind Nginx, with GitHub Actions CI, automated testing, health validation, controlled releases, rollback procedures, and React/TypeScript frontend integration.
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EARLIER EXPERIENCE

EMITEK — Lanham, MD

ETL Engineer / SQL Server DBA | July 2021–August 2022

- Built and supported Python and SSIS ETL pipelines integrating SQL Server, structured files, XML, APIs, and other business data sources.
 - Developed Python and Pandas data-processing workflows and improved ETL/SQL performance through query tuning, indexing, execution-plan analysis, and database optimization.
 - Automated scheduling and monitoring with Python, SQL Server Agent, and T-SQL while troubleshooting production pipeline failures, API issues, and data inconsistencies.
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EDUCATION

Bachelor of Science, Information Technology—BSIT to MSIT Pathway | 2026

Western Governors University

Associate Degree, Computer Science / Information Systems | 2021

Montgomery College

CERTIFICATIONS

Claude Certified Associate — Foundations

Claude Certified Developer—In Progress

Claude Certified Architect—In Progress

CompTIA Security+